

RETAIL CUSTOMER BEHAVIOR ANALYTICS

DASHBOARD REPORT

End-to-End Data Analysis using Python, PostgreSQL & Power BI

By -

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Technologies Used

Python • Pandas • PostgreSQL • SQL • Power BI • Jupyter Notebook

Project Type

End-to-End Data Analytics Project

GitHub Repository

<https://github.com/rakshit1504/Retail-Customer-Behavior-Analysis>

EXECUTIVE SUMMARY

Modern retail organizations generate enormous amounts of transactional data through both online and offline customer purchases. While this information contains valuable insights into customer preferences and purchasing behaviour, it often remains underutilized without systematic analysis. Extracting meaningful patterns from such data enables businesses to make informed decisions related to customer engagement, inventory planning, promotional campaigns, and revenue optimization.

This project presents a complete customer behavior analytics pipeline developed using Python, PostgreSQL, and Power BI. Beginning with a raw retail transaction dataset, the project demonstrates the complete analytical workflow including data cleaning, preprocessing, exploratory data analysis, business-oriented SQL querying, and interactive dashboard development.

Python and the **Pandas** library were used to inspect the dataset, handle preprocessing tasks, engineer additional features such as customer age groups, and prepare the data for further analysis. **PostgreSQL** was then utilized to answer practical business questions through SQL, enabling deeper insights into customer spending patterns, subscription behaviour, product performance, seasonal trends, payment preferences, and customer satisfaction. Finally, an interactive **Power BI** dashboard was designed to present the most important performance indicators and business metrics in a visually intuitive format.

Rather than focusing only on visualization, the project emphasizes analytical thinking by translating business requirements into measurable Key Performance Indicators (KPIs) and SQL-based business questions. The final dashboard enables users to quickly evaluate customer demographics, purchasing behaviour, category-wise sales, revenue distribution, and customer subscription patterns.

Overall, this project demonstrates the complete lifecycle of a modern data analytics solution, integrating data preprocessing, structured querying, visualization, and business interpretation into a single workflow suitable for practical retail analytics applications.

1. INTRODUCTION

Data has become one of the most valuable assets for modern businesses. Every customer transaction contributes valuable information that can help organizations better understand customer preferences, evaluate business performance, and identify opportunities for growth. However, raw transactional data alone provides limited value unless it is transformed into actionable business insights through systematic analysis.

Retail organizations continuously collect information regarding customer demographics, purchase history, product categories, payment methods, shipping preferences, promotional offers, and customer feedback. Analyzing these variables collectively enables businesses to optimize pricing strategies, improve customer retention, enhance inventory planning, and personalize marketing campaigns.

The objective of this project is to demonstrate an end-to-end data analytics workflow using widely adopted industry tools. Starting from raw retail transaction data, the project performs data preprocessing using Python, executes analytical SQL queries within PostgreSQL, and finally presents business insights through an interactive Power BI dashboard.

Unlike traditional reports that only summarize numerical findings, this project emphasizes business-oriented analysis by answering practical questions that organizations commonly encounter during decision-making processes.

The completed solution provides an accessible and interactive platform for understanding customer behaviour, enabling business stakeholders to identify patterns and monitor key performance indicators efficiently.

2. BUSINESS PROBLEM

Retail businesses rely heavily on customer purchasing data to evaluate performance and formulate strategic decisions. Despite collecting large amounts of transactional information, organizations often struggle to derive meaningful insights due to fragmented datasets and limited analytical processes.

Several important business questions remain unanswered without structured analysis, including:

- Which customer segments contribute the highest revenue?
- Do loyalty or subscription programs influence customer spending?

- Which product categories generate the highest sales?
- How effective are promotional discounts?
- Which age groups contribute the most to overall revenue?
- Are repeat customers more likely to subscribe?
- Which payment methods are preferred by customers?
- How does customer satisfaction vary across product categories?

Answering these questions requires a systematic approach involving data cleaning, exploratory analysis, SQL-based querying, and dashboard visualization. This project addresses these challenges by transforming raw customer transaction data into meaningful business intelligence.

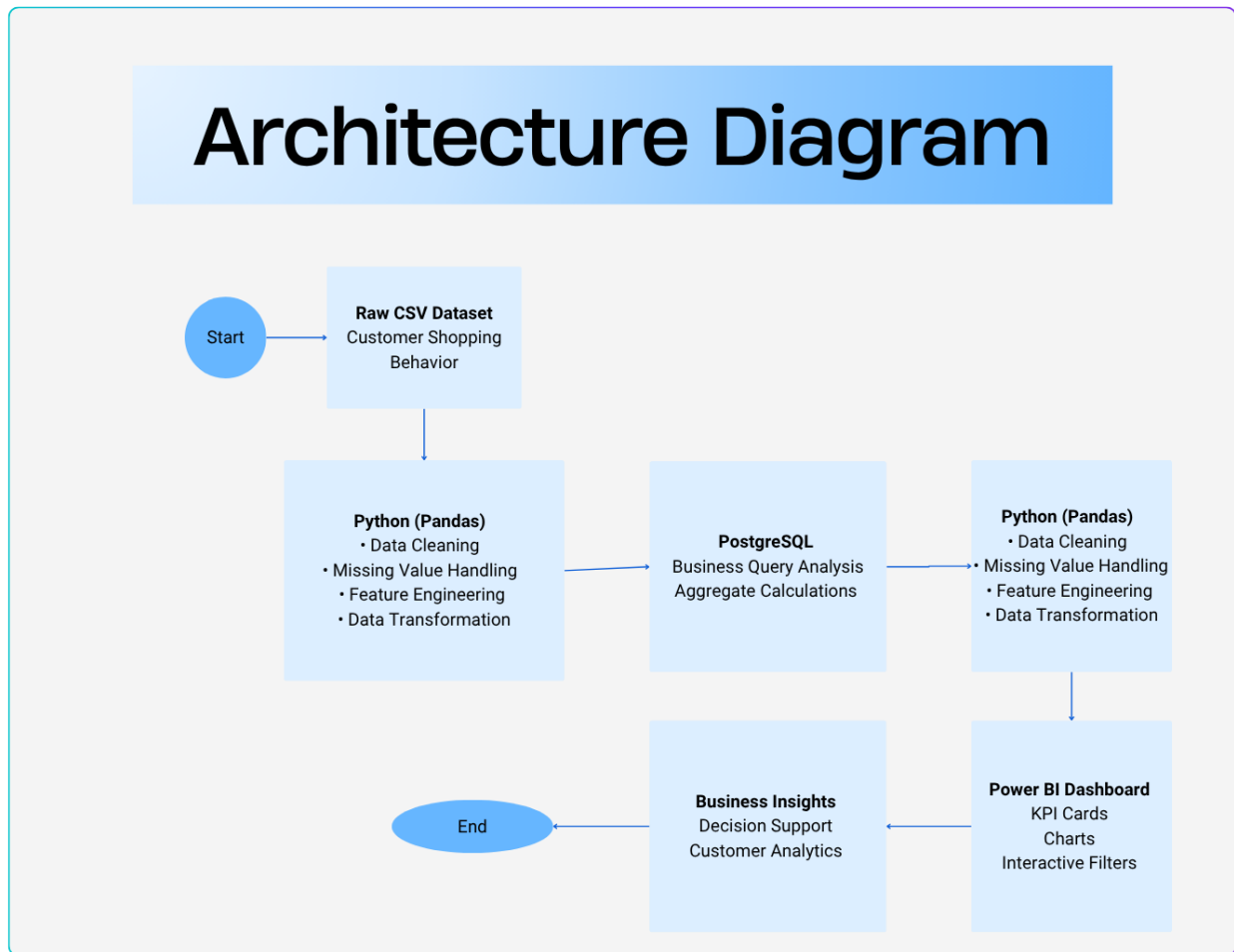
3. PROJECT OBJECTIVES

The primary objective of this project is to analyze retail customer purchasing behaviour and generate actionable business insights through an end-to-end analytics pipeline.

The specific objectives include:

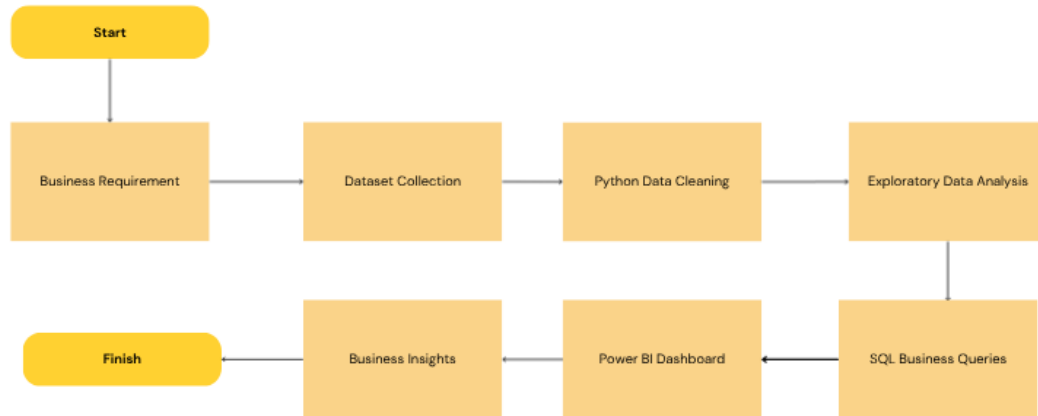
- Perform data cleaning and preprocessing using Python and Pandas.
- Examine the dataset for inconsistencies, missing values, and data quality issues.
- Engineer additional analytical features to simplify business reporting.
- Conduct exploratory data analysis to understand customer demographics and purchasing trends.
- Develop SQL queries to answer business-oriented analytical questions.
- Measure customer spending behaviour across different demographic groups.
- Evaluate the impact of subscriptions, discounts, shipping methods, and seasonal purchasing patterns.
- Design an interactive Power BI dashboard for visualizing key business metrics.
- Present analytical findings in a structured and business-friendly format.

4. PROJECT ARCHITECTURE



5. PROJECT WORKFLOW

Workflow Diagram



6. DATASET DESCRIPTION

6.1 Dataset Overview

The foundation of any analytical project lies in the quality and relevance of the dataset being analyzed. For this project, a retail customer shopping behavior dataset was used, containing transactional information collected from customer purchases across multiple product categories.

The dataset captures both demographic information and purchase-related attributes, enabling analysis from multiple business perspectives. Each record represents an individual customer transaction and includes variables associated with customer characteristics, purchasing decisions, shipping preferences, discounts, subscriptions, payment methods, and product reviews.

The diversity of features within the dataset makes it suitable for exploratory analysis, business intelligence reporting, and dashboard development.

Dataset Summary

Property	Details
Dataset Type	Retail Customer Transactions

File Format	CSV
Number of Records	Approximately 3900
Number of Features	18
Tools Used	Python, PostgreSQL, Power BI
Domain	Retail Analytics

6.2 Dataset Attributes

The dataset consists of the following important variables:

Column	Description
Customer ID	Unique identifier assigned to every customer
Age	Customer's age
Gender	Male or Female
Item Purchased	Name of the purchased product
Category	Product category
Purchase Amount	Amount spent by the customer
Location	Customer location
Size	Product size
Color	Product color
Season	Season during which the purchase occurred
Review Rating	Customer rating for the purchased item
Subscription Status	Whether the customer has an active subscription
Shipping Type	Delivery option selected
Discount Applied	Indicates whether a discount was used
Promo Code Used	Whether a promotional code was applied

Previous Purchases	Number of earlier purchases made by the customer
Payment Method	Payment option used
Frequency of Purchases	Customer purchase frequency

6.3 Why This Dataset?

This dataset was selected because it combines demographic information with transactional and behavioral attributes, allowing multiple aspects of customer behavior to be explored within a single analysis.

Unlike datasets that focus only on sales, this dataset includes customer subscriptions, promotional offers, product ratings, shipping methods, and historical purchasing activity. These additional variables make it possible to answer business-oriented questions that closely resemble real-world retail analytics scenarios.

7. TECHNOLOGY STACK

A combination of programming, database, and visualization tools was used to complete the project. Each technology served a specific purpose within the analytics workflow.

Technology	Purpose
Python	Data preprocessing and analysis
Pandas	Data manipulation and cleaning
PostgreSQL	Business query execution
SQL	Analytical query writing
Power BI	Dashboard development
Jupyter Notebook	Interactive development environment

8. DATA PREPARATION AND CLEANING

Before performing any analytical operations, the dataset was carefully examined to ensure data quality and consistency. Proper preprocessing is essential because inaccurate or incomplete data can significantly affect analytical outcomes.

The preprocessing stage was performed entirely using Python and the Pandas library within Jupyter Notebook.

The following workflow was followed.

Step 1: Importing Required Libraries

The analysis began by importing the required Python libraries.

Primary libraries included:

- Pandas
- NumPy (if used)
- Matplotlib (if used)
- Seaborn (if used)

These libraries provide functionality for data loading, preprocessing, statistical analysis, and visualization.

Figure 8.1:

Jupyter customer_behavior Last Checkpoint: 2 hours ago

File Edit View Run Kernel Settings Help Trusted

JupyterLab Python 3 (ipykernel)

```
[3]: import pandas as pd

df = pd.read_csv('customer_shopping_behavior_amlanmohanty.csv')
```

```
[4]: df.head()
```

	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
5	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	Yes	Yes	14	Venmo	Fortnightly
9	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	Yes	Yes	2	Cash	Fortnightly
10	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping	Yes	Yes	23	Credit Card	Weekly
11	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Air	Yes	Yes	49	PayPal	Weekly
15	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping	Yes	Yes	31	PayPal	Annually

Step 2: Loading the Dataset

The CSV dataset was imported into a Pandas DataFrame for analysis.

After loading, the first few records were inspected to verify successful import and understand the dataset structure.

The following inspections were performed:

- Previewing initial records
- Checking total rows and columns
- Understanding feature names

Step 3: Understanding the Dataset

To understand the structure of the dataset, descriptive information was generated using built-in Pandas functions.

The following properties were examined:

- Number of rows
- Number of columns

- Data types
- Memory usage
- Presence of missing values

This preliminary inspection helped identify potential preprocessing requirements before further analysis.

Figure 8.3:

The screenshot shows a Jupyter Notebook window titled 'customer_behavior' with a 'Last Checkpoint: 2 hours ago' status. The interface includes a menu bar (File, Edit, View, Run, Kernel, Settings, Help) and a toolbar with icons for saving, adding, deleting, and running code. The code cell [5] contains the command `df.info()`. The output displays the DataFrame's structure, including the number of entries (3900), the total number of columns (18), and a detailed breakdown of each column's data type and non-null count. The memory usage is reported as 548.6+ KB.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Customer ID                          3900 non-null   int64
1   Age                                  3900 non-null   int64
2   Gender                              3900 non-null   object
3   Item Purchased                       3900 non-null   object
4   Category                             3900 non-null   object
5   Purchase Amount (USD)                3900 non-null   int64
6   Location                             3900 non-null   object
7   Size                                 3900 non-null   object
8   Color                                3900 non-null   object
9   Season                               3900 non-null   object
10  Review Rating                        3863 non-null   float64
11  Subscription Status                  3900 non-null   object
12  Shipping Type                        3900 non-null   object
13  Discount Applied                     3900 non-null   object
14  Promo Code Used                      3900 non-null   object
15  Previous Purchases                   3900 non-null   int64
16  Payment Method                       3900 non-null   object
17  Frequency of Purchases               3900 non-null   object
dtypes: float64(1), int64(4), object(13)
memory usage: 548.6+ KB
```

Step 4: Missing Value Analysis

One of the first preprocessing tasks involved checking the dataset for incomplete or missing values.

Missing values can introduce bias into analytical results and therefore must be identified before conducting statistical analysis.

Each column was examined individually to determine whether any missing observations existed. Where required, suitable preprocessing techniques were applied to ensure data consistency.

Figure 8.3:

```
[6]: df.isnull().sum()

[6]: Customer ID      0
     Age             0
     Gender           0
     Item Purchased   0
     Category         0
     Purchase Amount (USD) 0
     Location         0
     Size             0
     Color            0
     Season           0
     Review Rating    37
     Subscription Status 0
     Shipping Type    0
     Discount Applied 0
     Promo Code Used  0
     Previous Purchases 0
     Payment Method   0
     Frequency of Purchases 0
     dtype: int64
```

Step 5: Data Cleaning

After identifying inconsistencies, the dataset underwent cleaning to improve its quality.

The cleaning process included:

- Handling missing values
- Verifying numerical columns
- Standardizing categorical entries
- Removing unnecessary inconsistencies
- Preparing the dataset for SQL analysis

These operations ensured that subsequent analysis was performed on reliable and consistent data.

9. FEATURE ENGINEERING

Raw datasets often require additional features that simplify business reporting.

Instead of repeatedly writing complex SQL conditions, new analytical variables were generated during preprocessing.

One such feature created during this project was **Age Group**.

Customers were categorized into meaningful demographic groups, allowing revenue and purchasing behavior to be analyzed more effectively across age segments.

The generated age groups included categories such as:

- Teen
- Young Adult
- Adult
- Senior

This transformation simplified both SQL queries and Power BI visualizations while improving the interpretability of analytical results.

Figure 9.1: *creation of the age_group column*

```
[10]: # create a new column age_group
labels = ['Young Adult', 'Adult', 'Middle-aged', 'Senior']
df['age_group'] = pd.qcut(df['age'], q=4, labels = labels)
df[['age', 'age_group']].head(10)
```

```
[10]:
```

	age	age_group
0	55	Middle-aged
1	19	Young Adult
2	50	Middle-aged
3	21	Young Adult
4	45	Middle-aged
5	46	Middle-aged
6	63	Senior
7	27	Young Adult
8	26	Young Adult
9	57	Middle-aged

10. EXPLORATORY DATA ANALYSIS

Once preprocessing was completed, exploratory data analysis (EDA) was performed to gain an initial understanding of customer behavior.

EDA focuses on summarizing important characteristics of the dataset before applying advanced analytical techniques.

The analysis aimed to answer questions such as:

- How are customers distributed across different age groups?
- Which product categories appear most frequently?
- How are purchase amounts distributed?
- What is the distribution of customer ratings?
- Are subscribed customers common?

- Which payment methods are frequently used?

Rather than drawing final conclusions, exploratory analysis serves as the foundation for identifying trends that can later be validated using SQL queries and dashboard visualizations.

Initial Observations

The exploratory analysis indicated that customer purchases were distributed across multiple product categories with relatively balanced representation.

Purchase amounts varied considerably among customers, suggesting the presence of both low-value and high-value transactions.

Customer ratings generally remained within the higher end of the rating scale, indicating positive overall customer satisfaction.

Subscription status, discount usage, shipping preferences, and purchase frequency displayed sufficient variation to support meaningful business analysis in later stages of the project.

11. SQL-BASED BUSINESS ANALYSIS

Following data preprocessing, the cleaned dataset was imported into PostgreSQL for business analysis. SQL was used to answer practical questions that retail organizations commonly encounter while evaluating customer behaviour and sales performance.

The analysis focuses on customer demographics, purchasing behaviour, subscriptions, discounts, product performance, and revenue contribution. Each query was designed to address a specific business objective rather than simply retrieve data.

Query 1: Revenue Generated by Male vs Female Customers

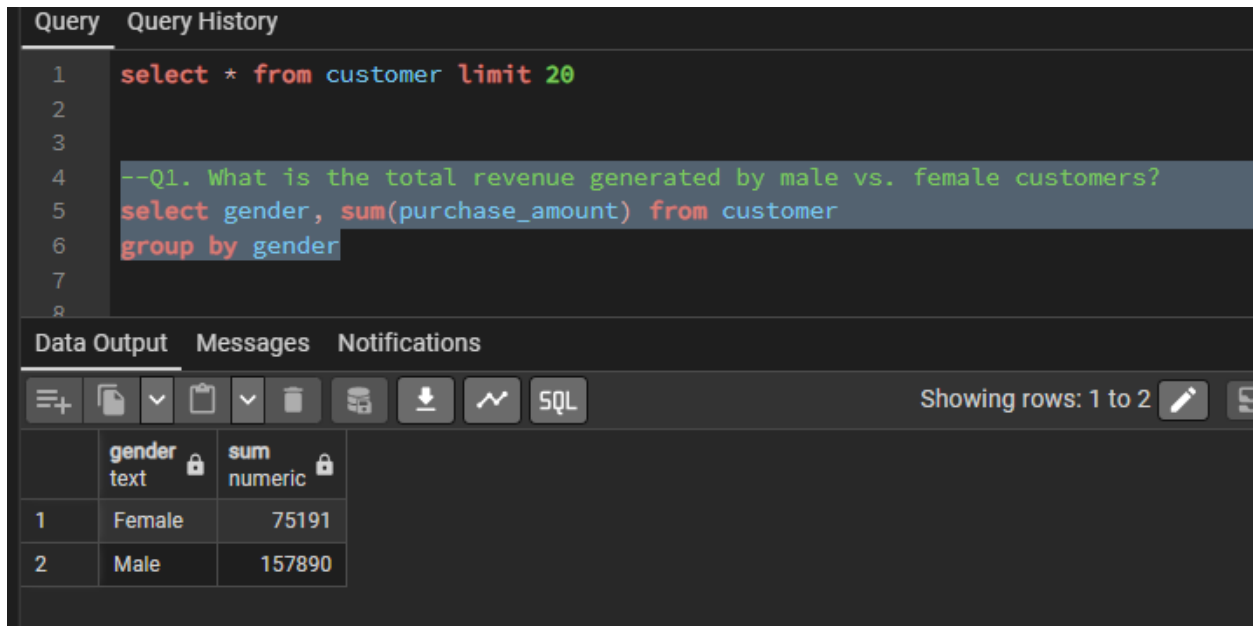
Business Objective

To compare the total revenue generated by male and female customers and understand purchasing behaviour across customer demographics.

SQL Approach

The dataset was grouped by gender, and the purchase amount was aggregated using the SUM() function.

Figure 11.1: SQL query and output screenshot



The screenshot shows a SQL IDE interface. The top panel, titled 'Query', contains a SQL query. The bottom panel, titled 'Data Output', displays the results of the query in a table format. The table has two columns: 'gender' (text) and 'sum' (numeric). The results show two rows: one for 'Female' with a sum of 75191, and one for 'Male' with a sum of 157890.

```
1 select * from customer limit 20
2
3
4 --Q1. What is the total revenue generated by male vs. female customers?
5 select gender, sum(purchase_amount) from customer
6 group by gender
7
8
```

	gender text	sum numeric
1	Female	75191
2	Male	157890

Business Interpretation

This analysis provides an overview of revenue contribution by gender. Such information can help retailers evaluate whether customer engagement or promotional strategies differ across demographic groups and whether targeted marketing initiatives may be beneficial.

Query 2: High-Value Customers Purchasing with Discounts

Business Objective

Identify customers who received discounts but still spent more than the overall average purchase amount.

SQL Approach

A subquery was used to calculate the average purchase amount. Customers with applied discounts whose purchases exceeded this average were then filtered.

Figure 11.2:

Query	Query History
6	group by gender
7	
8	
9	--Q2. Which customers used a discount but still spent more than the average purchase amount?
10	select customer_id, purchase_amount from customer
11	where discount_applied = 'Yes'
12	and purchase_amount > (select avg(purchase_amount) from customer)
13	

Data Output	Messages	Notifications																																	
Showing rows: 1 to 839 Page No: 1																																			
<table> <thead> <tr> <th></th><th>customer_id bigint</th><th>purchase_amount bigint</th></tr> </thead> <tbody> <tr><td>1</td><td>2</td><td>64</td></tr> <tr><td>2</td><td>3</td><td>73</td></tr> <tr><td>3</td><td>4</td><td>90</td></tr> <tr><td>4</td><td>7</td><td>85</td></tr> <tr><td>5</td><td>9</td><td>97</td></tr> <tr><td>6</td><td>12</td><td>68</td></tr> <tr><td>7</td><td>13</td><td>72</td></tr> <tr><td>8</td><td>16</td><td>81</td></tr> <tr><td>9</td><td>20</td><td>90</td></tr> <tr><td>10</td><td>22</td><td>62</td></tr> </tbody> </table>		customer_id bigint	purchase_amount bigint	1	2	64	2	3	73	3	4	90	4	7	85	5	9	97	6	12	68	7	13	72	8	16	81	9	20	90	10	22	62		
	customer_id bigint	purchase_amount bigint																																	
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4	7	85																																	
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6	12	68																																	
7	13	72																																	
8	16	81																																	
9	20	90																																	
10	22	62																																	

Business Interpretation

Contrary to the assumption that discounts primarily attract low-value purchases, this analysis identifies premium customers who also utilize promotional offers. This insight helps businesses evaluate the effectiveness of discount campaigns across different customer segments.

Query 3: Highest Rated Products

Business Objective

Determine which products consistently receive the highest customer satisfaction ratings.

SQL Approach

Products were grouped by item name, and their average review rating was calculated before selecting the top five.

Figure 11.3:

Query

Query History

13

14

15

16

17

18

19

```

-- Q3. Which are the top 5 products with the highest average review rating?
select item_purchased, avg(review_rating) from customer
group by item_purchased
order by avg(review_rating) desc
limit 5

```

Data Output

Messages

Notifications

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SQL

Showing rows: 1 to 5

✎

	item_purchased	avg
	text	double precision
1	Gloves	3.8614285714285725
2	Sandals	3.8443750000000003
3	Boots	3.8187500000000005
4	Hat	3.8012987012987005
5	Skirt	3.784810126582278

Business Interpretation

Highly rated products indicate strong customer satisfaction and can guide inventory planning, premium marketing campaigns, and future product recommendations.

Query 4: Standard vs Express Shipping Analysis

Business Objective

Compare customer spending between Standard and Express shipping options.

SQL Approach

Average purchase amounts were calculated separately for Standard and Express shipping methods.

Figure 11.4:

```
19
20 --Q4. Compare the average Purchase Amounts between Standard and Express Shipping.
21 select shipping_type, avg(purchase_amount) from customer
22 where shipping_type in ('Standard', 'Express')
23 group by shipping_type;
24
```

Data Output Messages Notifications

Showing rows: 1 to 2

	shipping_type text	avg numeric
1	Standard	58.4602446483180428
2	Express	60.4752321981424149

Business Interpretation

Understanding whether customers selecting faster shipping also spend more helps retailers optimize delivery strategies and premium shipping offerings.

Query 5: Subscription Impact on Customer Spending

Business Objective

Evaluate whether subscribed customers contribute greater revenue than non-subscribed customers.

SQL Approach

Customer counts, average purchase amounts, and total revenue were calculated for subscribed and non-subscribed customers.

Figure 11.5:

Query

Query History

25

--Q5. Do subscribed customers spend more? Compare average spend and total revenue

26

--between subscribers and non-subscribers.

27

SELECT subscription_status,

28

COUNT(customer_id) AS total_customers,

29

ROUND(AVG(purchase_amount),2) AS avg_spend,

30

ROUND(SUM(purchase_amount),2) AS total_revenue

31

FROM customer

32

GROUP BY subscription_status

33

ORDER BY total_revenue,avg_spend DESC;

Data Output

Messages

Notifications

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SQL

Showing rows: 1 to 2

Page

	subscription_status text	total_customers bigint	avg_spend numeric	total_revenue numeric
1	Yes	1053	59.49	62645.00
2	No	2847	59.87	170436.00

Business Interpretation

Subscription programs often represent customer loyalty. Comparing spending behaviour between these groups provides valuable insights into the effectiveness of subscription-based engagement strategies.

Query 6: Products Most Frequently Purchased with Discounts

Business Objective

Identify which products are most commonly purchased using discounts.

SQL Approach

The percentage of discounted purchases was calculated for each product using conditional aggregation.

Figure 11.6:

Query

Query History

36

--Q6. Which 5 products have the highest percentage of purchases with discounts applied?

37

select item_purchased, 100*sum(case when discount_applied = 'Yes' then 1 else 0 end)/count(*) as percent

38

from customer

39

group by item_purchased

40

order by percent desc

41

limit 5

42

Data Output

Messages

Notifications

Showing rows: 1 to 5

Page No: 1 of 1

	item_purchased text	percent bigint
1	Hat	50
2	Sneakers	49
3	Coat	49
4	Sweater	48
5	Pants	47

Business Interpretation

This analysis highlights products that rely heavily on promotional pricing, helping businesses evaluate discount dependency and pricing strategies.

Query 7: Subscription Behaviour Among Repeat Customers

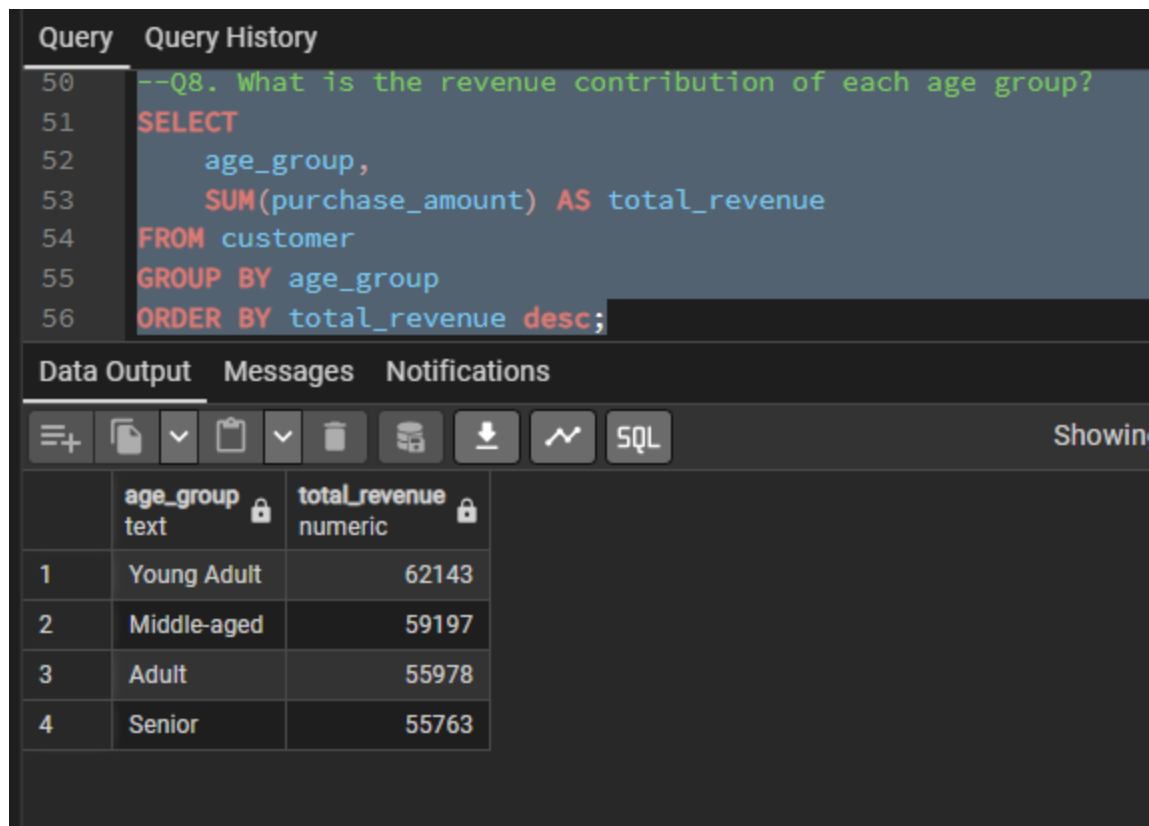
Business Objective

Determine whether customers with a strong purchasing history are more likely to subscribe.

SQL Approach

Customers having more than five previous purchases were filtered and grouped by subscription status.

Figure 11.7:



Business Interpretation

Revenue distribution across age groups enables businesses to identify their most valuable customer segments and tailor marketing campaigns accordingly.

Query 9: Revenue by Payment Method

Business Objective

Determine customer payment preferences and evaluate revenue generated through different payment methods.

SQL Approach

Orders, total revenue, and average purchase amounts were grouped by payment method.

Figure 11.9:

Query

Query History

59

-- Q9. Which payment methods generate the highest revenue?

60

SELECT

61

payment_method,

62

COUNT(*) AS total_orders,

63

ROUND(SUM(purchase_amount),2) AS total_revenue,

64

ROUND(AVG(purchase_amount),2) AS avg_order_value

65

FROM customer

66

GROUP BY payment_method

67

ORDER BY total_revenue DESC;

68

Data Output

Messages

Notifications

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SQL

Showing rows: 1 to

	payment_method text	total_orders bigint	total_revenue numeric	avg_order_value numeric
1	Credit Card	671	40310.00	60.07
2	PayPal	677	40109.00	59.25
3	Cash	670	40002.00	59.70
4	Debit Card	636	38742.00	60.92
5	Venmo	634	37374.00	58.95
6	Bank Transfer	612	36544.00	59.71

Business Interpretation

Understanding preferred payment methods assists retailers in optimizing payment partnerships, improving checkout experiences, and introducing targeted promotional offers.

Query 10: Seasonal Sales Analysis

Business Objective

Identify seasonal trends in customer purchasing behaviour.

SQL Approach

Orders and revenue were aggregated according to season.

Figure 11.10:

Query		Query History		
68				
69		-- Q10. Which season contributes the highest sales?		
70		SELECT		
71		season,		
72		COUNT(*) AS total_orders,		
73		ROUND(SUM(purchase_amount),2) AS total_revenue,		
74		ROUND(AVG(purchase_amount),2) AS avg_purchase		
75		FROM customer		
76		GROUP BY season		
77		ORDER BY total_revenue DESC;		
Data Output		Messages		
		Showing results		
	season	total_orders	total_revenue	avg_purchase
	text	bigint	numeric	numeric
1	Fall	975	60018.00	61.56
2	Spring	999	58679.00	58.74
3	Winter	971	58607.00	60.36
4	Summer	955	55777.00	58.41

Business Interpretation

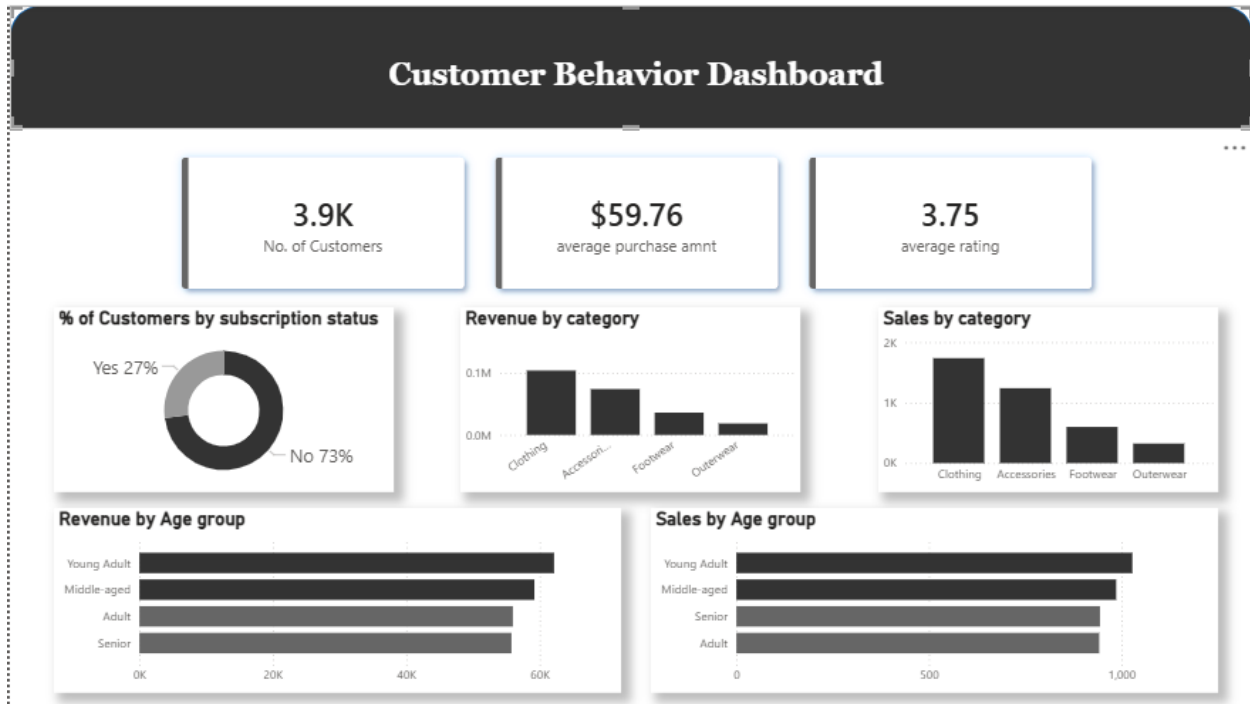
Seasonal demand analysis supports inventory planning, promotional scheduling, and workforce allocation throughout the year.

12. POWER BI DASHBOARD

After completing the SQL-based analysis, the findings were consolidated into an interactive dashboard developed using Microsoft Power BI. The objective of the dashboard was to provide business stakeholders with a concise overview of customer behaviour through intuitive visualizations and key performance indicators.

The dashboard was designed with a clean layout, allowing users to quickly identify revenue trends, customer distribution, purchasing behaviour, and category-wise performance. Rather than displaying raw data, the dashboard emphasizes actionable insights that support business decision-making.

Figure 12.1:



Dashboard Components

The dashboard consists of the following key elements:

- **KPI Cards:** Display total customers, average purchase amount, and average review rating to provide a quick summary of business performance.
- **Revenue by Category:** Highlights the contribution of each product category to total revenue, enabling easy identification of high-performing categories.
- **Customer Distribution:** Visualizes customer demographics based on gender and subscription status.
- **Revenue by Age Group:** Illustrates which demographic segments contribute most significantly to sales.
- **Category-wise Sales:** Compares sales across product categories to identify purchasing trends.

The dashboard transforms analytical findings into an interactive business intelligence solution, enabling stakeholders to explore data from multiple perspectives without requiring technical expertise.

13. KEY BUSINESS INSIGHTS

The analysis produced several meaningful observations regarding customer purchasing behaviour and retail performance:

- Customer spending patterns vary across demographic groups, indicating opportunities for more targeted marketing strategies.
- Subscription programs appear to influence purchasing behaviour and contribute significantly to overall business revenue.
- Several customers continue to make high-value purchases even when promotional discounts are available, suggesting that discounts are not exclusively used by low-spending customers.
- Revenue is concentrated within a limited number of product categories, highlighting areas of strong customer demand.
- Customer satisfaction remains consistently high across many products, indicating positive overall shopping experiences.
- Seasonal trends suggest opportunities for optimizing inventory planning and promotional campaigns.
- Repeat customers demonstrate characteristics associated with long-term customer loyalty, emphasizing the importance of customer retention initiatives.

14. LEARNING OUTCOMES

This project provided valuable practical experience in applying the complete data analytics lifecycle to a real-world retail dataset. Through the project, I strengthened my understanding of data preprocessing, exploratory analysis, SQL-based business querying, and interactive dashboard development.

Working with Python and Pandas improved my ability to clean and transform raw datasets into analysis-ready formats. Using PostgreSQL reinforced my SQL skills by translating business requirements into structured queries that generated meaningful insights. Designing the Power BI dashboard enhanced my understanding of data visualization principles and effective communication of analytical findings.

Beyond technical skills, the project emphasized the importance of framing analytical work around business questions rather than focusing solely on tools or code. This experience strengthened my ability to connect technical analysis with practical business decision-making.

15. CONCLUSION

This project demonstrates a complete end-to-end customer analytics workflow using Python, PostgreSQL, and Power BI. Starting from raw transactional data, the project progressed through data cleaning, exploratory analysis, SQL-driven business intelligence, and dashboard visualization to produce actionable insights into customer purchasing behaviour.

By integrating multiple technologies within a single analytical pipeline, the project illustrates how raw retail data can be transformed into meaningful information that supports strategic decision-making. The resulting dashboard provides stakeholders with an accessible platform for monitoring customer demographics, purchasing trends, subscription behaviour, and product performance.